Reliable Smartphone-Based Detection of Counterfeit Bangladeshi Banknotes: Multi-View Deep Learning with Out-of-Distribution Rejection and a Benchmark Audit

Debobrato Biswas and Imran Mahmud

Abstract—Counterfeit high-value banknotes harm retailers and individuals who cannot afford dedicated verification hardware. This paper presents a smartphone-based authentication system for Bangladeshi Tk 500 and Tk 1000 notes and evaluates it on JaalTaka, a public benchmark of 1,390 notes (802 genuine and 588 counterfeit notes seized by law enforcement) with six photographs per note. The system fine-tunes an ImageNet-pretrained EfficientNet-B0, fuses the predictions of up to six photographs of a note, and returns a three-way verdict (genuine, uncertain, or counterfeit) with an adjustable strictness threshold. An embedding-space Mahalanobis-plus-residual test rejects photographs that do not resemble the supported notes, and Grad-CAM heatmaps show which regions drove each decision. On 208 held-out test notes, the system reaches 98.08% note-level accuracy (95% CI 95.16–99.25%) and an ROC-AUC of 0.994, and runs at 37 ms per photograph on a desktop CPU. An audit of the benchmark shows why these figures must be read with care. Pixel-level matching groups the 1,390 notes into only 846 distinct photograph sets: 116 test notes have an exact or augmented copy in the training or validation split, and 12 duplicate groups carry both labels. All four test errors fall in these label-conflict groups. On the 92 test notes without any copy in training, the system makes no errors, but so do a 1-nearest-neighbour classifier and a linear probe on frozen ImageNet features. The benchmark is therefore saturated under note-level splitting. We release the duplicate groups and recommend group-aware splits and counterfeit-family hold-out tests.

Index Terms—Banknote authentication, Bangladeshi Taka, counterfeit detection, data leakage, EfficientNet, explainable AI, Grad-CAM, out-of-distribution detection, transfer learning.

Word count: 3,097 words (main text); abstract: 246 words.

Plain-Language Summary

Fake Tk 500 and Tk 1000 notes are hard to spot by eye. We built a tool that checks a banknote from one to six ordinary phone photographs. It answers “genuine”, “counterfeit”, or “uncertain, please check by hand”, refuses pictures that do not show a supported note, and highlights the parts of the note that influenced its answer. On a public collection of banknote photographs, it was correct for 98% of the test notes. When we examined that collection closely, we found many repeated photographs, some labelled both genuine and fake, and these repeats explained every mistake. The high score therefore does not prove that the tool will catch new kinds of fakes. We share the list of repeated photographs and suggest fairer ways to test such tools.

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Word count: 3,097 words in the main text (Sections I–VI, excluding the abstract, tables, figure captions, algorithm, back matter, and references); abstract: 246 words; 9 figures; 8 tables.

I. Introduction

COUNTERFEIT banknotes undermine trust in cash, which remains the main means of payment for small retailers and much of the population in Bangladesh. The Tk 500 and Tk 1000 notes are the preferred targets of counterfeiters because a single accepted forgery causes a large loss. Banks use sorting machines with ultraviolet, infrared, and magnetic sensors, but shopkeepers and ordinary citizens rely on visual and tactile checks of the watermark, security thread, and intaglio print, which are slow and unreliable for untrained users. A smartphone camera is the one sensor almost everyone carries, which makes image-based authentication an attractive complement to manual checks.

Earlier counterfeit detectors relied on hand-crafted features [1]–[3], often under controlled scanner or multispectral imaging [4]. Convolutional neural networks (CNNs) trained on smartphone photographs have since detected forgeries of several currencies [5], [6]. For Bangladeshi currency, most work addressed denomination recognition [7], [8], until the JaalTaka dataset [9] released smartphone photographs of genuine notes and of counterfeits seized by the Rapid Action Battalion (RAB). Its authors reported test accuracies of 97.4–98.9% for five ImageNet CNNs.

High benchmark accuracy is necessary, but not sufficient, for a tool that people will trust with money. A deployable system must also (i) behave sensibly when the input is not a supported note, (ii) admit uncertainty, (iii) explain its decision, and (iv) be evaluated in a way that predicts its behaviour on unseen notes. The last point is easy to get wrong: deep networks exploit shortcuts [10], and duplicated samples across training and test splits silently inflate reported accuracy [11].

This paper addresses these requirements with the following contributions.

1) A complete, open, CPU-real-time authentication pipeline for Tk 500 and Tk 1000 notes (Fig. 1). It combines an EfficientNet-B0 [12] classifier, fusion of up to six photographs of one note, a three-way decision rule with an adjustable strictness threshold, an embedding-space out-of-distribution (OOD) gate, and Grad-CAM [13] explanations.
2) A note-disjoint evaluation on JaalTaka that reports confidence intervals, per-photograph and per-note results, threshold behaviour, OOD rejection rates, and comparisons with frozen-feature baselines.
3) A pixel-level duplicate audit of the benchmark. It shows that most test notes have copies in the training split, that several duplicate groups carry contradictory labels and cause every test error of our system, and that the benchmark is saturated once duplicates are controlled. An occlusion test further indicates that part of the discriminative signal lies outside the note's central region.

Sections II–VII cover related work, the dataset and protocol, the proposed system, the experiments and audit, the discussion, and the conclusion.

II. Related Work

A. Hand-Crafted Counterfeit Detection

Yeh et al. [1] classified luminance histograms of Taiwanese banknotes with multiple-kernel support vector machines (SVMs). Berenguel et al. [2] analysed scanned background printing texture, which differs between the intaglio and offset printing of genuine notes and the inkjet or laser printing of most forgeries. Baek et al. [4] exploited the multispectral response of inks. Uddin et al. [3] inspected individual security features of Bangladeshi notes, and Dittimi and Suen [14] built a mobile application. These methods are interpretable and light, but depend on controlled imaging and on features tuned for each denomination.

B. Deep Learning for Banknotes

Transfer learning from ImageNet [15], [16] allows CNNs to be trained with the few thousand images typically available for banknotes. Pham et al. detected fake banknotes in smartphone images to assist visually impaired users [5] and later studied cross-dataset detection [6]. For Bangladeshi notes, neural networks and lightweight CNNs were used for denomination recognition [7], [8]. The JaalTaka authors [9] benchmarked counterfeit detection with DenseNet121 [17], ResNet50 [18], VGG16 [19], MobileNetV2 [20], and InceptionV3 [21], and transfer learning was also applied in [22]. Table I summarises these studies; none evaluates OOD inputs, uncertainty, or the duplicate structure of the data.

C. Reliability: OOD Detection, Explanation, and Leakage

A two-class classifier assigns one of its classes to any input, including unrelated objects. The maximum softmax probability is a useful but weak OOD signal [23]; Lee et al. [24] obtained stronger detection with Mahalanobis distances on intermediate features, an approach we adapt to a low-dimensional subspace. Network probabilities are often miscalibrated [25], which motivates an explicit "uncertain" outcome. Grad-CAM [13] highlights the image regions that support a class. Finally, networks solve benchmarks through unintended shortcuts [10], and leakage between training and test data, including duplicated samples, inflates published results [11]. Our audit (Section V-H) applies these lessons to counterfeit detection.

III. Dataset and Evaluation Protocol

A. The JaalTaka Dataset

JaalTaka [9], [26] contains photographs of 1,390 Bangladeshi notes: 802 genuine notes (398 Tk 500 and 404 Tk 1000) collected from three national banks, and 588 counterfeit notes (320 Tk 500 and 268 Tk 1000) supplied by RAB for research. Each note was photographed six times with one of four smartphones (Samsung Galaxy A50 and M31, Redmi Note 8 and Note 9 Pro), giving 8,340 images. The six photographs cover different segments of the front and back of the note (Fig. 2). The dataset description states that 649 original notes (502 genuine and 147 counterfeit) were expanded to 1,390 by augmentation. We resized every image so that its longer side is 1,024 pixels; the data are distributed under the CC BY 4.0 licence.

B. Note-Disjoint Split

Because the six photographs of a note are strongly correlated, we split the data by note, never by photograph. Within each class, notes were shuffled with a fixed seed (42) and assigned 70%/15%/15% to training, validation, and test (Table II). The validation split is used for model selection and OOD calibration; the test split is used once. As Section V-H shows, a note-disjoint split is not sufficient on this dataset, because distinct note identifiers can share photographs.

IV. Proposed System

Fig. 1 shows the processing chain. Each of one to six photographs of a note is classified independently; photographs that fail the OOD gate are discarded, the remaining probabilities are averaged, and a three-way decision is made. A Grad-CAM heatmap accompanies every photograph.

A. Classifier

We use EfficientNet-B0 [12], which offers a good trade-off between accuracy and computation on CPUs and mobile devices, initialised with ImageNet weights [15]. The convolutional feature extractor $f(\cdot)$ maps an input x to a $1280 \times 9 \times 9$ feature map for 288×288 inputs; global average pooling yields the embedding $\mathbf{e} = f(x) \in \mathbb{R}^{1280}$.

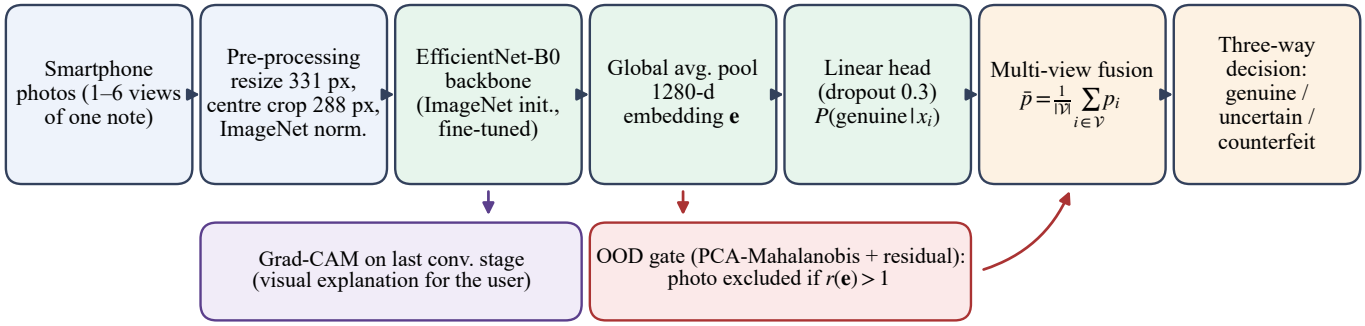


Fig. 1. Overview of the proposed system. Each photograph x_i of a note is classified by a fine-tuned EfficientNet-B0. Photographs whose embedding $\mathbf{e}$ fails the OOD test are excluded, the genuine probabilities of the remaining views are averaged, and the note is labelled genuine, uncertain, or counterfeit. Grad-CAM explains each per-photograph decision.

TABLE I
Representative Banknote Authentication Studies

Study	Currency	Imaging	Approach and task
Yeh et al. [1] (2011)	Taiwan dollar	Image sensor	Luminance histograms + multiple-kernel SVM; counterfeit detection
Berenguel et al. [2] (2016)	Banknotes	Flatbed scanner	Background printing-texture features + SVM; counterfeit detection
Uddin et al. [3] (2016)	BDT	Camera images	Image-based analysis of security features; counterfeit detection
Baek et al. [4] (2018)	Banknotes	Multispectral	Spectral response of inks; counterfeit detection
Dittimi and Suen [14] (2018)	Banknotes	Smartphone	Mobile application; counterfeit detection
Pham et al. [5] (2020)	Banknotes	Smartphone, visible light	CNN; fake-banknote detection for visually impaired users
Linkon et al. [8] (2020)	BDT	Camera images	Lightweight CNN + transfer learning; denomination recognition
Pham et al. [6] (2022)	USD, EUR, KRW, JOD	Smartphone	CNN; fake detection across datasets
Piyas et al. [9] (2025)	BDT 500/1000	Smartphone (4 models)	JaalTaka dataset; five ImageNet CNN baselines (best 98.9%)
Tabassum et al. [22] (2025)	BDT	Images	Transfer learning; counterfeit detection
This work	BDT 500/1000	Smartphone, 1–6 views	EfficientNet-B0, multi-view fusion, three-way decision, OOD gate, Grad-CAM, duplicate audit

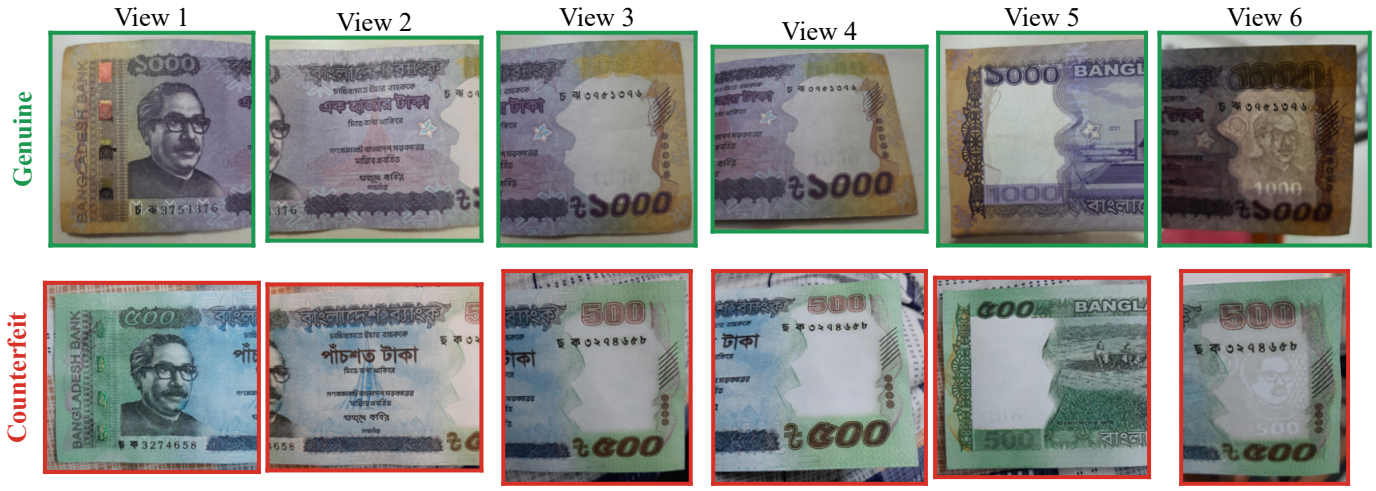


Fig. 2. The six photographs of one genuine Tk 1000 note (top) and one counterfeit Tk 500 note (bottom) from the JaalTaka test split. Each view shows a different segment of the note; the colour, lighting, and background vary with the phone and the session.

The ImageNet classifier is replaced by dropout ($p = 0.3$) and a linear layer with two outputs, so that The model has 4.01 M parameters.

$$p(x) = \mathrm{softmax}(\mathbf{We} + \mathbf{b})_{\text{genuine}}\,. \qquad (1)$$

TABLE II
Note-Disjoint Split of JaalTaka (Seed 42)

	Notes			Photographs	
Split	Genuine	Counterf.	Total	Genuine	Counterf.
Training	562	412	974	3,372	2,472
Validation	120	88	208	720	528
Test	120	88	208	720	528
Total	802	588	1,390	4,812	3,528

TABLE III
Training Configuration

Setting	Value
Backbone	EfficientNet-B0, ImageNet-1k weights
Frozen layers	Stem + first 3 MBConv stages
Head	Dropout 0.3, linear $1280 \to 2$
Input size	288×288
Optimiser	AdamW, weight decay 10^{-4}
Learning rate	One-cycle, peak 3×10^{-4}, 15% warm-up
Batch size / epochs	12 / 6 (487 steps per epoch)
Loss	Weighted cross-entropy, label smoothing 0.05
Model selection	Max. validation (accuracy + AUC)
Random seed	42 (split, initialisation, shuffling)

B. Training

Training images are augmented with random resized crops (scale 0.35–1.0, aspect ratio 0.75–1.33, output 288×288), rotations of up to $\pm 10°$, colour jitter (brightness and contrast ± 0.4, saturation ± 0.3, hue ± 0.04), and, with probability 0.2, Gaussian blur, so that the network relies on print and texture rather than lighting or framing. At evaluation, the shorter side is resized to 331 pixels and the central 288×288 region is kept, which removes most of the background.

The stem and the first three MBConv stages, which compute generic filters, are frozen together with their ImageNet batch-normalisation statistics. The rest of the network is fine-tuned with the class-weighted, label-smoothed [21] cross-entropy

$$\mathcal{L} = -\sum_{c \in \{0,1\}} w_c \, q_c \log \hat{y}_c, \quad q_c = (1-\varepsilon)\mathbb{1}[c = y] + \tfrac{\varepsilon}{2}, \tag{2}$$

where $\hat{y}$ is the softmax output, $\varepsilon = 0.05$, and $w_c = N/(2N_c)$ compensates for the 562:412 class imbalance. We use AdamW [27] and a one-cycle learning-rate schedule [28]; Table III lists the settings. After each epoch, the weights with the highest sum of validation accuracy and AUC are kept. Validation accuracy rose from 98.80% after the first epoch to 99.04%, and validation AUC from 0.9919 to 0.9976 after the sixth, which was selected. Training took about 55 min on an Intel Core i5-12400 CPU without a GPU, using PyTorch [29].

C. Multi-View Fusion and Three-Way Decision

Let $\mathcal{V}$ be the set of photographs of one note that pass the OOD gate (Section IV-D). The note-level genuine probability is the mean

$$\bar{p} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} p(x_i). \tag{3}$$

Averaging needs no extra training and accepts any number of views; it is the late-fusion analogue of view pooling in multi-view CNNs [30]. Given a strictness threshold $\tau \in [0.5, 1)$, the verdict is

$$\text{verdict} = \begin{cases} \text{not supported} & \text{if } \mathcal{V} = \emptyset, \\ \text{genuine} & \text{if } \bar{p} \geq \tau, \\ \text{counterfeit} & \text{if } \bar{p} < 0.5, \\ \text{uncertain} & \text{otherwise.} \end{cases} \tag{4}$$

The rule reflects the asymmetric cost of errors: accepting a counterfeit causes a direct loss, whereas "uncertain" only asks the user to check the note by hand or to add photographs. The default is $\tau = 0.7$; users can choose 0.6, 0.7, or 0.85, or set any value in the interface.

D. Out-of-Distribution Gate

Following the feature-space approach of Lee et al. [24], we model the embeddings of the training photographs. Let $\boldsymbol{\mu}$ be their mean and $\mathbf{U}_k \in \mathbb{R}^{1280 \times k}$ the first $k = 64$ principal directions, with variances $\sigma_1^2, \ldots, \sigma_k^2$. For an embedding $\mathbf{e}$ with $\mathbf{z} = \mathbf{U}_k^\top(\mathbf{e} - \boldsymbol{\mu})$, we compute the Mahalanobis distance inside the subspace and the energy outside it:

$$m(\mathbf{e}) = \sum_{j=1}^{k} \frac{z_j^2}{\sigma_j^2}, \qquad \rho(\mathbf{e}) = \|(\mathbf{e} - \boldsymbol{\mu}) - \mathbf{U}_k \mathbf{z}\|_2^2 . \tag{5}$$

Both thresholds, θ_m and θ_ρ, are set to the 99.5th percentile of the validation photographs, and the OOD ratio is $r(\mathbf{e}) = \max\{m(\mathbf{e})/\theta_m,\ \rho(\mathbf{e})/\theta_\rho\}$. A photograph is accepted when $r(\mathbf{e}) \leq 1$ and when its texture, measured as the variance of the Laplacian of a 512×512 grey-scale version, is at least 30, which rejects blank and heavily blurred images. The residual term catches embeddings that leave the banknote subspace, which the in-subspace distance alone cannot detect.

E. Visual Explanation

For each photograph, Grad-CAM [13] is computed on the last convolutional feature map $A \in \mathbb{R}^{1280 \times 9 \times 9}$ for the predicted class c:

$$\alpha_k = \frac{1}{81} \sum_{u,v} \frac{\partial y^c}{\partial A_{uv}^k}, \qquad L^c = \mathrm{ReLU}\Big(\sum_k \alpha_k A^k \Big), \tag{6}$$

where y^c is the logit of class c. The map is normalised, upsampled to the analysed region, and overlaid on the photograph.

Algorithm 1 Authentication of One Banknote

Require: Photographs $x_1, \ldots, x_n$ of one note ($1 \le n \le 6$), threshold τ
Ensure: Verdict, $\bar{p}$, and one heatmap per photograph

$\mathcal{V} \leftarrow \emptyset$
for $i = 1$ to n **do**
  $\mathbf{e}_i \leftarrow f(\text{crop}(x_i))$; $p_i \leftarrow$ Eq. (1)
  $H_i \leftarrow$ Grad-CAM of the predicted class, Eq. (6)
  if $r(\mathbf{e}_i) \le 1$ and $\text{texture}(x_i) \ge 30$ **then**
    $\mathcal{V} \leftarrow \mathcal{V} \cup \{i\}$
  end if
end for
if $\mathcal{V} = \emptyset$ **then**
  return not supported
end if
$\bar{p} \leftarrow \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} p_i$
return verdict from Eq. (4), $\bar{p}$, $\{H_i\}$

F. Inference Procedure and Deployment

Algorithm 1 summarises inference. The deployed detector averages two crops per photograph (the central crop of Section IV-B and the full frame resized to 288×288); all results in Section V use the central crop only, which matches the training-time evaluation. The model runs in 37 ms per photograph (median of 30 runs, batch size one, six CPU threads). Flask and Streamlit interfaces show the verdict, per-photograph probabilities, heatmaps, and the effect of changing τ.

V. Experiments and Results

A. Metrics

Genuine is the positive class. We report accuracy; the counterfeit detection rate (specificity, the share of counterfeits rejected); the genuine acceptance rate (recall); and the area under the ROC curve (AUC) [31]. Accuracy is given with a 95% Wilson score interval [32]. Results are reported per photograph (n = 1,248) and per note after fusing all six photographs (n = 208). Unless stated otherwise, decisions use the threshold 0.5, i.e., the two-way limit of Eq. (4).

B. Main Results

Table IV and Fig. 3 summarise the test results. Per photograph, the system reaches 98.08% accuracy and an AUC of 0.990. Per note, it rejects 86 of 88 counterfeits and accepts 118 of 120 genuine notes (98.08% accuracy, AUC 0.994). Fig. 4 shows that both ROC curves rise almost vertically; at the default operating point, 2.3% of counterfeit notes are accepted.

The per-note confusion matrix is exactly the per-photograph matrix divided by six, and each view position alone reaches 98.08%: every error occurs on all six photographs of one of four specific notes, which Section V-H examines.

TABLE IV
Test Results of the Proposed System (Threshold 0.5)

Metric	Per photograph	Per note (6 views)
Samples (genuine / counterf.)	1,248 (720 / 528)	208 (120 / 88)
Accuracy (%)	98.08	98.08
95% Wilson CI (%)	97.15–98.70	95.16–99.25
Counterfeit detection (%)	97.73	97.73 (86/88)
Genuine acceptance (%)	98.33	98.33 (118/120)
Balanced accuracy (%)	98.03	98.03
F1, genuine class	0.983	0.983
ROC-AUC	0.990	0.994

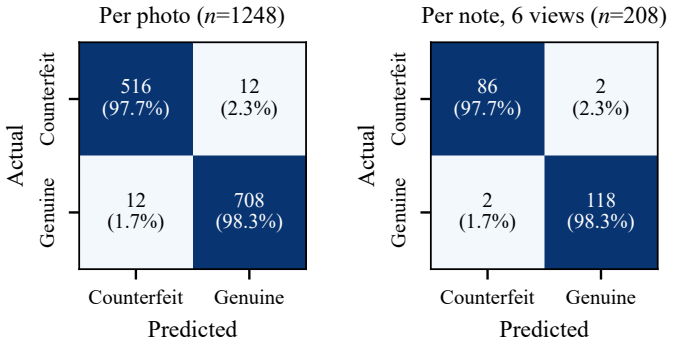


Fig. 3. Confusion matrices on the test split, per photograph (left) and per note after fusing six photographs (right). Percentages are row-normalised.

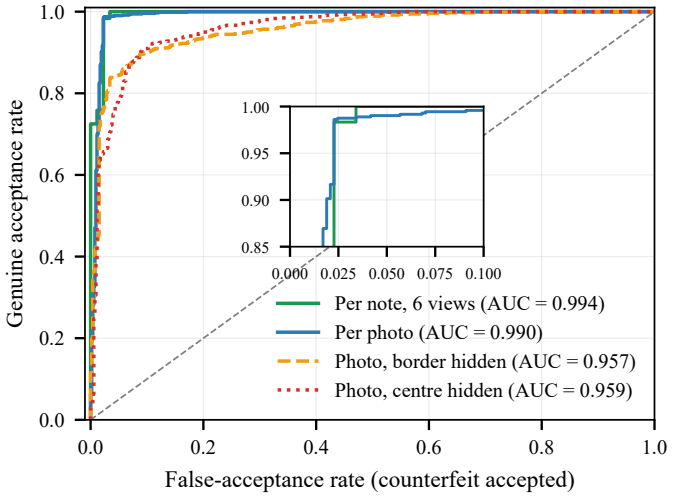


Fig. 4. ROC curves on the test split for per-note fusion, single photographs, and the two occlusion diagnostics of Section V-G. The inset magnifies the low-false-acceptance region.

C. Comparison With Baselines

To isolate the effect of fine-tuning, we evaluated two classifiers on frozen ImageNet EfficientNet-B0 embeddings (same split and crop): a cosine 1-nearest-neighbour (1-NN) classifier and a class-balanced logistic-regression probe, with note-level fusion as in Eq. (3). Table V shows that fine-tuning improves per-photograph accuracy by 1.4–1.6 points and AUC by 0.009–0.023. At note level, however, the linear probe matches the accuracy of the fine-tuned model, and even the 1-NN classifier errs on only six notes. For reference, the JaalTaka authors reported 97.4–98.9%

TABLE V
Comparison With Frozen-Feature Baselines on the Test Split

	Per photograph		Per note	
Method	Acc. (%)	AUC	Acc. (%)	AUC
1-NN, frozen ImageNet features	96.71	0.967	97.12	0.979
Linear probe, frozen features	96.47	0.981	98.08	0.983
Fine-tuned EfficientNet-B0 (ours)	98.08	0.990	98.08	0.994

TABLE VI
Effect of the Number of Fused Photographs (Test Notes)

Photographs k	1	2	3	4	5	6
Subsets per note	6	15	20	15	6	1
Accuracy (%)	98.08	98.08	98.08	98.08	98.08	98.08
Mean AUC	0.9900	0.9912	0.9917	0.9925	0.9930	0.9939

for five fine-tuned CNNs [9]; their split protocol differs from ours, so those figures are not directly comparable.

D. Multi-View Fusion and the Strictness Threshold

To measure the value of additional photographs, we evaluated Eq. (3) on every subset of k views of each test note ($\binom{6}{k}$ subsets per k). Table VI shows that accuracy is the same for every subset, again because all errors are note-level, whereas the mean AUC increases monotonically from 0.990 with one photograph to 0.994 with six: fusion makes the ranking of notes more reliable and pushes the scores of both classes further from the decision boundary.

Fig. 5(a) shows that the fused scores are strongly bimodal: all correctly classified genuine notes have $\bar{p} > 0.86$ and all correctly classified counterfeits $\bar{p} < 0.10$. Consequently, the three-way rule of Eq. (4) leaves the outcome unchanged for $\tau \leq 0.85$; at $\tau = 0.90$, one genuine note (0.8%) becomes "uncertain" (Fig. 5(b)). The two accepted counterfeits have $\bar{p} \approx 0.95$ and would only be caught with $\tau > 0.954$, which would send most genuine notes to manual inspection; as Section V-H shows, they are not ordinary misclassifications.

E. Out-of-Distribution Gate

Fig. 6 shows the distribution of the OOD ratio $r(\mathbf{e})$. On the test split, 4 of 1,248 photographs (0.32%) are rejected, against 0.72% on the calibration (validation) split; the median test ratio is 0.26, and no test note was declared "not supported". In a preliminary check, the gate rejected 100 of 105 photographs of other Bangladeshi denominations (Tk 2–200) taken from a separate collection, as well as blank, noise, and unrelated images. A systematic OOD benchmark with independently collected images remains future work.

F. Visual Explanations

Fig. 7 shows Grad-CAM heatmaps for the genuine test note with the highest fused score, the counterfeit note with the lowest, and one of the two accepted counterfeits. For correct decisions, the heatmaps concentrate on the portrait and on the denomination text and ornament panels, which carry the finest printed detail; on the counterfeit in row (b), the security band at the left border also contributes. The heatmaps of the wrongly accepted note in row (c) resemble those of a genuine note, as expected if, as shown next, an identical copy appears in training labelled genuine.

TABLE VII
Duplicate Audit of JaalTaka (Pixel NCC > 0.98 Over Six Views)

Quantity	Value
Notes / distinct photograph sets (groups)	1,390 / 846
Groups of size 1 / 2 / 3	449 / 250 / 147
Notes sharing their photographs with another note	941 (67.7%)
Groups with only counterfeit / only genuine labels	289 / 545
Groups containing both labels	12 (31 notes)
Test notes with a copy in training or validation	116 of 208 (55.8%)
counterfeit / genuine	60 of 88 / 56 of 120
Test notes in label-conflict groups	6
Test errors explained by label conflicts	4 of 4

G. Shortcut Diagnostic

To test whether the model relies on the note or on its surroundings, we blacked out either the central region of each test input (keeping a 20% border) or the border (keeping the centre), as illustrated in Fig. 8, and evaluated the unchanged model. Hiding the border reduces accuracy to 88.94% (AUC 0.957). Hiding the centre reduces accuracy to 75.48%, because most genuine photographs are then rejected (genuine acceptance 58.47%), but the AUC remains 0.959 and 98.67% of counterfeits are still recognised. The 20% border alone thus suffices to rank most photographs correctly. Part of the border is the note itself (margins, security band, serial number), but it often also shows the table or hand beneath it, and genuine and counterfeit notes came from different sources (banks and RAB), likely photographed in different sessions. We cannot rule out that background, lighting, or the phone contribute to the decision, a classic shortcut risk [10].

H. Benchmark Audit: Duplicates and Label Conflicts

The perfect agreement between the photographs of a note, the strong frozen baselines, and the documented augmentation led us to search for duplicated notes. For every pair of notes, we computed the normalised cross-correlation (NCC) between 48×48 grey-scale thumbnails of corresponding views, which is invariant to brightness and contrast changes, averaged it over the six views, and took the maximum over horizontal and vertical flips. The distribution of the best match is clearly bimodal (Fig. 9(a)): for the validation and test notes, matches below the gap reach at most 0.963, whereas matches above it are at least 0.997. We therefore grouped notes whose six views match with NCC > 0.98 (connected components). Table VII summarises the findings.

Three findings stand out. First, the 1,390 notes contain only 846 distinct photograph sets, so the augmented

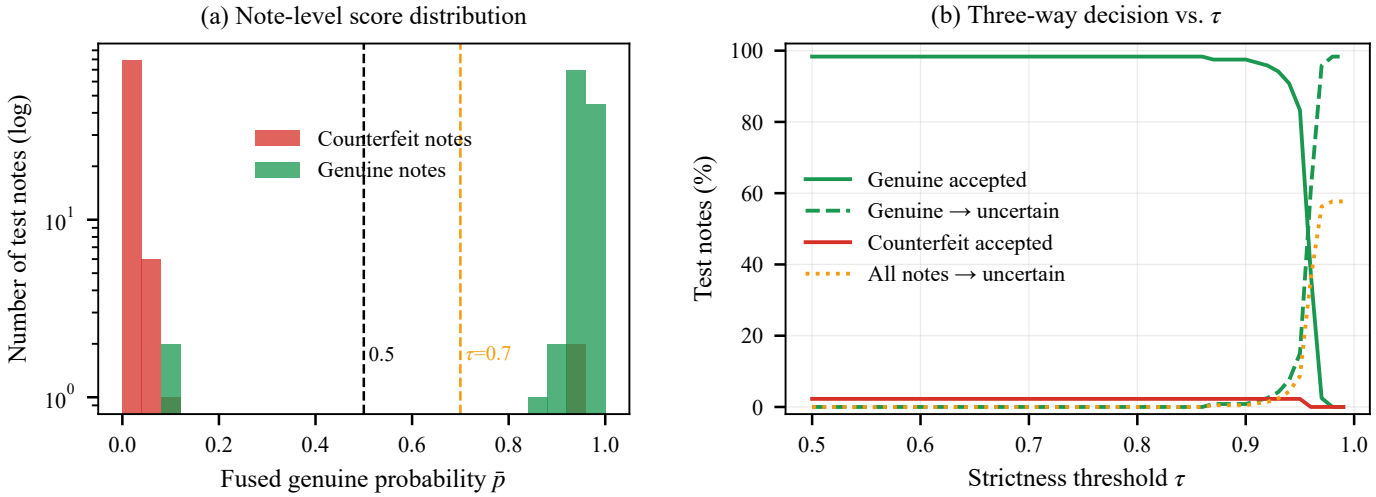


Fig. 5. (a) Distribution of the fused genuine probability $\bar{p}$ of the 208 test notes (log scale); dashed lines mark the counterfeit limit 0.5 and the default strictness $\tau = 0.7$. (b) Share of test notes accepted as genuine, sent to "uncertain", or wrongly accepted, as a function of the strictness threshold τ.

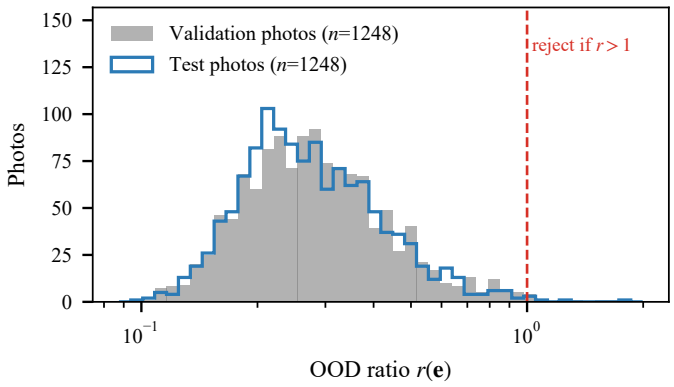


Fig. 6. OOD ratio $r(\mathbf{e})$ of validation photographs (used for calibration) and test photographs. Photographs to the right of the dashed line are excluded from the note-level decision.

TABLE VIII
Note-Level Accuracy (%) on Audited Test Subsets

Test subset	All	Without label conflicts	Clean (no copy)
Notes (genuine / counterf.)	208 (120/88)	202 (117/85)	92 (64/28)
1-NN, frozen features	97.12	100.0	100.0
Linear probe, frozen	98.08	100.0	100.0
Ours (fine-tuned)	98.08	100.0	100.0
95% Wilson CI	95.16–99.25	98.13–100	95.99–100

copies mentioned in the dataset description were stored as independent notes. A note-level split thus places copies of 55.8% of the test notes in the training or validation data. Second, 12 groups contain both genuine- and counterfeit-labelled copies of the same photographs (e.g., genuine notes 400–406 duplicate counterfeit notes 106–112 and 547–553, and counterfeit notes 400 and 403–406 duplicate genuine notes 802 and 1–4). At least one label in each of these groups must be wrong. The genuine-labelled notes 402 and 403, for instance, carry the serial number 1843834, which also appears on several counterfeit-labelled notes in the dataset. Third, all four test errors of our system are notes from these label-conflict groups, and in each case the model predicts the label of the copy it saw during training.

We therefore re-scored the system and the baselines on two subsets of the original test split (Table VIII). On the 92 clean test notes, whose duplicate group has no member in training or validation, all three methods are error-free. After removing only the six label-conflict notes, all three are also error-free on the remaining 202 notes. The 98.08% note-level accuracy is thus limited by the label noise of the benchmark, not by the model, and on correctly labelled notes the task is saturated even for a nearest-neighbour classifier on generic features.

VI. Discussion

A. What the Results Do and Do Not Show

Within JaalTaka, the system separates genuine and counterfeit notes perfectly once label conflicts are removed, with a lower 95% confidence bound of 96.0% on the 92 notes without a copy in training; its errors are fully explained, and it runs in real time on a CPU. These results do not, however, show that it will detect new counterfeits. The 588 counterfeit notes collapse into 289 counterfeit-only photograph sets, those we inspected share a few serial numbers, and they were collected separately from the genuine notes. A model, or even a 1-NN classifier, can therefore succeed by recognising known counterfeit batches or capture sessions, as the occlusion diagnostic suggests, rather than generic evidence of forgery. Saturation also means that the benchmark can no longer rank methods: differences between published CNNs on JaalTaka are smaller than the effect of its label noise.

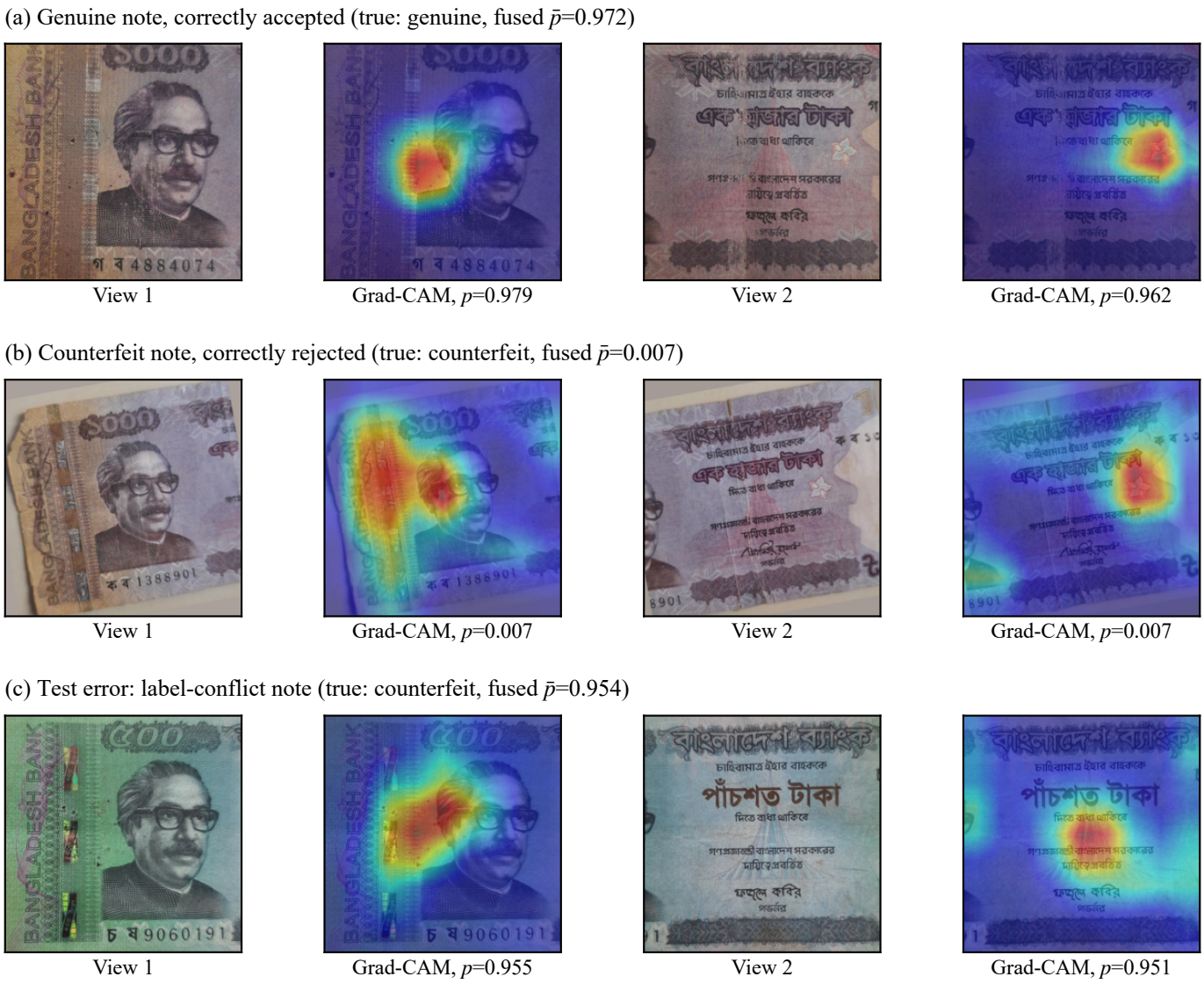


Fig. 7. Grad-CAM explanations on held-out test notes (views 1 and 2; red indicates high relevance). (a) Genuine note correctly accepted. (b) Counterfeit note correctly rejected. (c) Counterfeit note wrongly accepted; its photographs also occur in the training set under the genuine label (Section V-H).

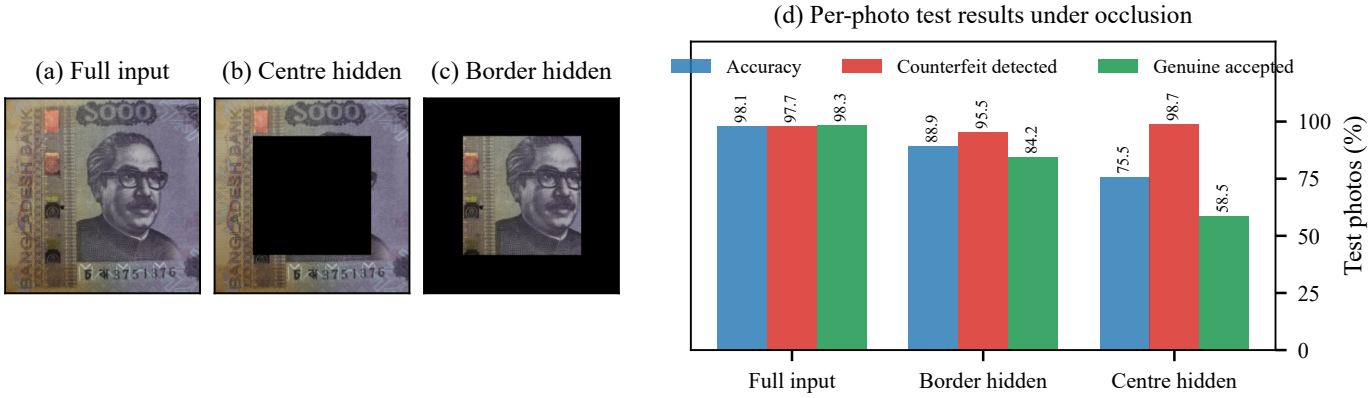


Fig. 8. Occlusion diagnostic. (a)–(c) Evaluation input, with the centre or the 20% border hidden. (d) Per-photograph test accuracy, counterfeit detection rate, and genuine acceptance rate of the unchanged model under each condition.

B. Recommendations for the Benchmark

For future work on JaalTaka and similar datasets, we recommend (i) removing or relabelling the 12 label-

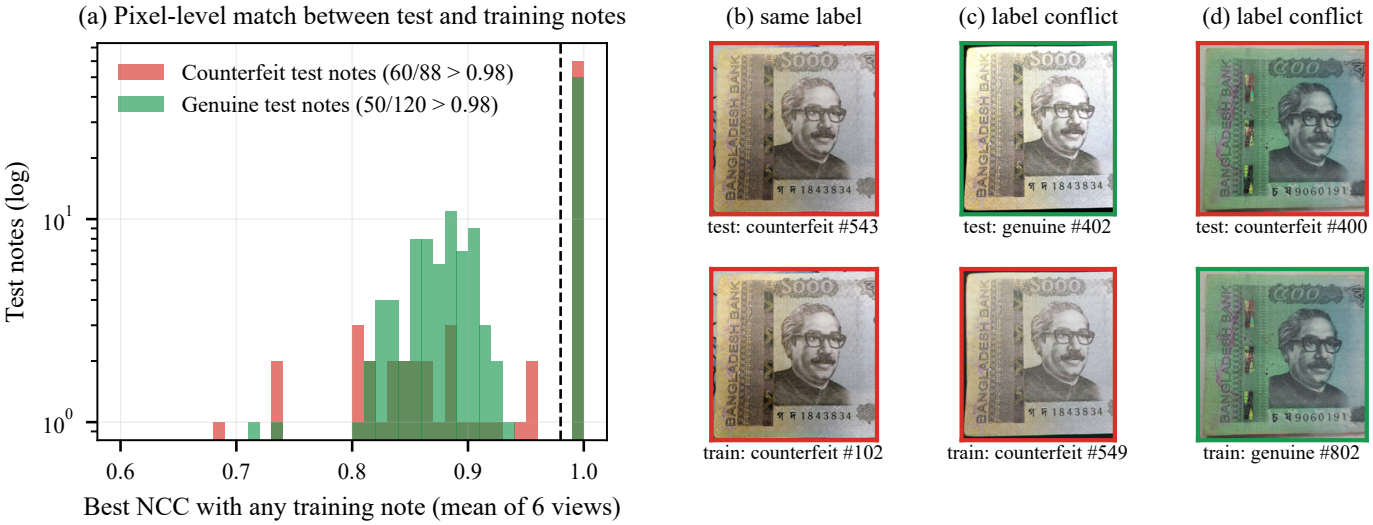


Fig. 9. Duplicate audit. (a) Best pixel-level match (mean NCC over the six views) between each test note and any training note; the dashed line marks the duplicate threshold 0.98. (b)–(d) First view of a test note (top) and its matching training note (bottom); frame colour gives the dataset label (green: genuine, red: counterfeit). (b) An augmented copy with the same label. (c), (d) Identical photographs stored under opposite labels.

conflict groups; (ii) splitting by duplicate group, not by note identifier, for which we release our group list; (iii) holding out entire counterfeit families (e.g., by serial number or seizure batch) and capture devices, to measure generalisation to unseen forgeries and phones; and (iv) reporting results with confidence intervals and simple frozen-feature baselines, which immediately reveal a saturated benchmark.

C. Limitations

The system covers only Tk 500 and Tk 1000; other notes are answered with "not supported". A visible-light photograph cannot capture ultraviolet features, magnetic ink, or intaglio relief, so the output is decision support, and suspicious notes should be confirmed at a bank. The training data contain only close-up phone photographs, so scans and distant photographs are mostly rejected by the OOD gate. The OOD evaluation on other denominations is preliminary, and the class probabilities have not been calibrated on an independent dataset [25]. Finally, we did not retrain the model on a group-aware split; doing so, together with a counterfeit-family hold-out, is the most important next step.

VII. Conclusion

We presented a reliability-oriented, CPU-real-time smartphone system for detecting counterfeit Tk 500 and Tk 1000 notes, combining multi-view fusion, a three-way decision, an OOD gate, and Grad-CAM explanations. It achieves 98.08% note-level accuracy and an AUC of 0.994 on a note-disjoint JaalTaka split, but our audit shows that this figure mainly reflects the benchmark: label conflicts among duplicated notes explain every test error, and on duplicate-free notes even simple baselines are error-free. Progress now depends less on new architectures than on evaluation data that separates counterfeit families, capture sessions, and devices.

Acknowledgment

The authors thank the Department of Software Engineering, Daffodil International University, for laboratory facilities and administrative support, and the creators of the JaalTaka dataset for making their data publicly available. This research received no external funding.

Author Contributions

D. Biswas: conceptualisation, software, experiments, formal analysis, and writing of the original draft. I. Mahmud: supervision, methodology review, and review and editing of the manuscript.

Conflict of Interest

The authors declare that they have no conflict of interest.

Ethics Statement

This study did not involve human participants, personal data, or animals. It used only photographs of banknotes from the publicly available JaalTaka dataset, distributed under the CC BY 4.0 licence [26]. Ethics approval and informed consent were therefore not required.

Generative AI Disclosure

The authors used an AI assistant to help write analysis and plotting scripts and to edit the manuscript text. The authors checked all code, results, and references and take full responsibility for the content of this paper.

Data and Code Availability

The JaalTaka dataset is publicly available from Mendeley Data under the CC BY 4.0 licence [26]. The trained model, the training and evaluation code, the scripts that reproduce every table and figure, and the list of duplicate groups found in the audit are available from the corresponding author and will be released in a public repository upon publication.

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